

HESHAN PRAMUDITHA DHARMASENA

BSc (Hons) IT Undergraduate | Data Science Specialization

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PROFESSIONAL SUMMARY

Third-year IT undergraduate specializing in Artificial Intelligence and Machine Learning, with strong foundations in linear algebra, probability, and statistics. Experienced in developing end-to-end ML systems, including computer vision models and Retrieval-Augmented Generation (RAG)-based NLP applications. Proficient in deploying scalable ML solutions using Python, Docker, and AWS. Seeking an AI/ML related internship to contribute to real-world systems and production-level applications.

SKILLS

Machine Learning & AI: PyTorch, Scikit-learn, Computer Vision (YOLOv8), NLP, RAG, LangChain, XGBoost, Neural Networks

Languages: Python, Java, C++, JavaScript, SQL

MLOps & Cloud: Docker, GitHub Actions (CI/CD), AWS (EC2, S3, ECR, Elastic Beanstalk), FastAPI, Flask

Data & Tools: NumPy, Pandas, Matplotlib, PostgreSQL, MySQL, MongoDB, ChromaDB

Web Development: Next.js, MERN Stack, Tailwind CSS

Soft Skills: Teamwork, Time Management, Critical Thinking, Problem Solving, Communication

PROJECTS

SOLIX.AI — AI-Powered Solar Energy Feasibility System | [GitHub](#)

Tech Stack: *PyTorch, YOLOv8, LangChain, RAG, FastAPI*

- Trained a YOLOv8 model on a custom Sri Lankan satellite dataset, achieving 80% mAP for rooftop detection.
- Developed a Retrieval-Augmented Generation (RAG) pipeline to generate responses based on technical energy documents.
- Engineered an end-to-end system integrating computer vision and NLP for real-world solar energy analysis.

End-to-End ML Framework with MLOps | [GitHub](#)

Tech Stack: *Scikit-learn, Docker, AWS, Flask, GitHub Actions*

- Designed a modular ML pipeline for data preprocessing, feature engineering, and model training.
- Implemented a CI/CD pipeline to automate testing and deployment using GitHub Actions.
- Containerized and deployed the application on AWS for scalable cloud access.
- Integrated logging and monitoring for system performance tracking.

MedAgent SL — Voice-Based Medical AI Assistant | [GitHub](#)

Tech Stack: *LangChain, RAG, Whisper, LLMs, Streamlit*

- Developed a low-latency medical Q&A system using RAG, improving response speed and answer accuracy through optimized retrieval and processing pipeline.
- Integrated speech-to-text and text-to-speech capabilities for real-time voice interaction.
- Designed a multimodal pipeline combining NLP and audio processing.

TrackNFix — Full-Stack AI-Powered Workshop Management System | [GitHub](#)

Tech Stack: *Next.js, Node.js, PostgreSQL, Docker, AWS*

- Developed a full-stack platform for vehicle lifecycle tracking, streamlining workshop operations and automating service workflows.
- Integrated an AI assistant for technical support and task automation.
- Deployed a scalable, cloud-based architecture using Docker and AWS.

EDUCATION

BSc (Hons) in Information Technology | Specialization: Data Science

University of Kelaniya, Sri Lanka | GPA: 3.54 / 4.00

2024 – Present

GCE Advanced Level — Physical Science Stream

Bandaranayake College, Gampaha, Sri Lanka

2020 – 2023

CERTIFICATIONS

- Neural Networks and Deep Learning — DeepLearning.AI
- Linear Algebra — DeepLearning.AI
- Probability & Statistics for Machine Learning — DeepLearning.AI
- Python for Data Science & AI — Coursera
- DevOps Foundation — LearnFi Academy

ACHIEVEMENTS & ACTIVITIES

- 4th Place — JuniorHack 7.0 Pre-Hackathon (University of Kelaniya)
- Technical Member — HackX & HackX Jr
- PR Coordinator — EvolveX 2.0 (Rotaract Club, University of Kelaniya)
- Technical Crew — TEDxUoK

EXTRACURRICULAR ACTIVITIES

Sports: Badminton (University Level), Football

Music: Violin (Provincial Level), Digital Music Production

REFERENCES

Mr. Sankatha Bamunuge

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